Introduction

The first half of the semester, consisting of Units 1–3, is an application of classical physics to understand the behavior of light. Starting with Unit 4, we begin to explore experiments that classical physics cannot explain, leading us to develop new models based on quantum mechanics. In Unit 4, we explore experimental results that suggest there are limitations with classical physics, and in Unit 5, we formalize quantum mechanics to explain these results.

Unit 4: Modern Physics

In this unit, we explore the failures of classical explanations for certain experimental phenomena, which will lead to the development of a *quantized* model of modern physics. We will begin with the development of the atomic model.

Generally, classical physics refers to the fields of Newtonian mechanics, electromagnetism, and thermodynamics.

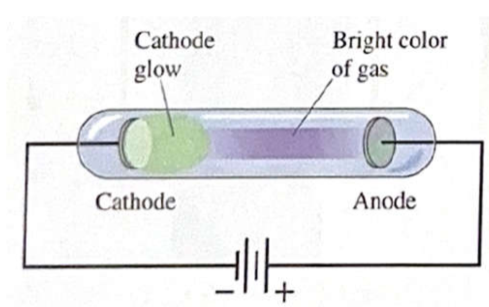
Evolution of Atomic Models

1800: Invention of the First Battery
When Alessandro Volta invented the first battery, electricity was thought to be a fluid that flowed through wires.

As a result of this discovery, scientists were able to perform *electrolysis*. By passing electricity through water, they were able to prove that water was made of two different elements, hydrogen and oxygen. Previously, water was thought to be basic and indivisible.

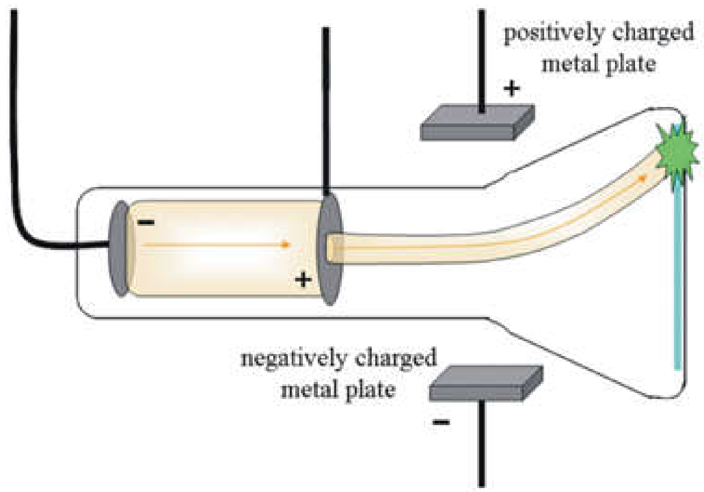
1808: Dalton Model
John Dalton argued that all matter is made up of atoms, which are indivisible and indestructible.

1820: Cathode Ray
When Michael Faraday tried to pass electricity through a gas, he observed that the current splits the gas into positively and negatively charged ions. The negative discharge always had a green glow, called a *cathode ray*.



This proved that atoms were not indivisible, as they could be split into charged particles.

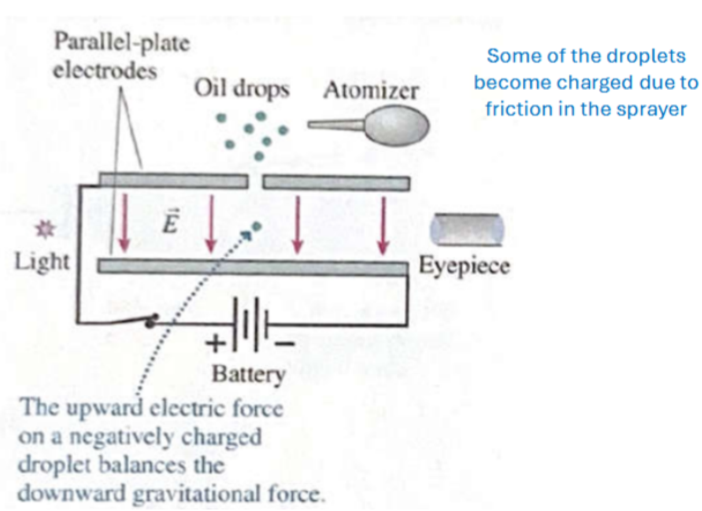
1888: Discovery of the Electron
J.J. Thomson discovered the electron by applying a magnetic field to the cathode ray. He observed that the cathode ray was deflected by the magnetic field, which suggested that it was made of negatively charged particles.



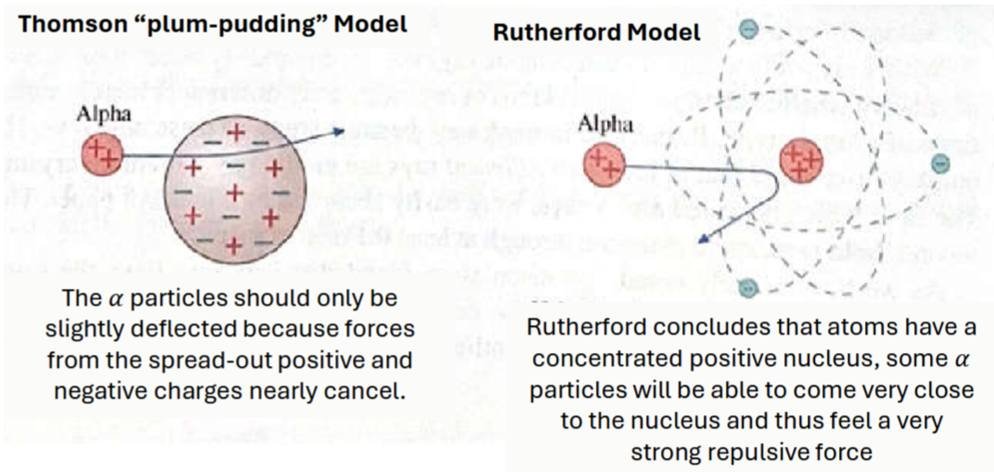
Thomson then proposes the **plum pudding atomic model**. That atoms are composed of *discrete* negatively charged electrons in a *continuous* positively charged medium.

1905: Miliken's Oil Drop Experiment
Robert Miliken finds that the charge on each varying oil drop is a multiple of 1.6×10^{-19} C. Thus, he proposes that the **charge of an electron** is $e = 1.6 \times 10^{-19}$ C.

Here, we realize that *charge is quantized*: $q = ne$, where $n = \pm 1, \pm 2, \pm 3, \ldots$.



1909: Gold Foil Experiment
Ernest Rutherford finds that the atoms are mostly empty space with a small, dense, positively charged nucleus. Thus, he proposes the **Rutherford atomic model**, where electrons orbit a small, dense positively charged nucleus.



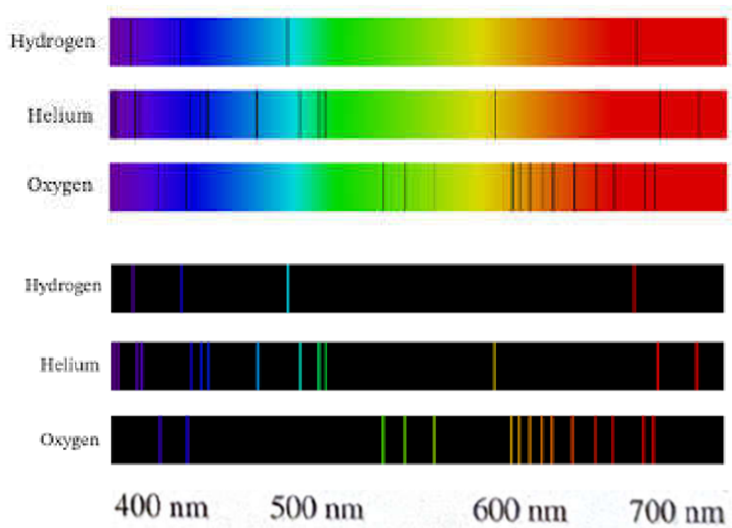
1913: Bohr Model
It would not be until Niels Bohr that we would have a quantized model of the atom. Bohr's atom is based on the Rutherford model, but with the addition of quantized energy levels.

1926: Schrodinger Model
Finally, Erwin Schrodinger develops the quantum mechanical model of the atom, which explains the quantized nature of the atom using quantum mechanics. Instead of discrete electrons orbiting the nucleus, Schrodinger's model describes electrons as a *cloud* of probability around the nucleus.

Generally, this development shows an intuition that we need to apply to our current understanding of classical physics to develop modern physics that explains the quantized nature of the universe. In the atomic model, we see through a series of experiments that classical assumptions of continuous charge and energy were incorrect. We will then continue this intuition through the discrete emission spectrum of hydrogen, which classical models of continuous light could not explain.

Emission and Absorption Spectra

We find that different elements emit and absorb light at discrete wavelengths. The **absorption spectrum** describes the wavelengths of light that are not reflected by an element after being exposed to a continuous spectrum of light. The **emission spectrum** describes the wavelengths of light that are emitted by an element after being excited by an external energy source.



Rydberg Formula

Experimentally derived, the Rydberg formula describes the wavelengths of light emitted by Hydrogen.

$$\boxed{\frac{1}{\lambda} = R_H\left(\frac{1}{m^2} - \frac{1}{n^2}\right), \quad R_H \approx 1.097 \times 10^7\,\mathrm{m}^{-1}}$$

For integers $n > m$ where n is the higher energy level and m is the lower energy level.

m	Series Name	Region
1	Lyman	Ultraviolet
2	Balmer	Visible
3	Paschen	Infrared
4	Brackett	Infrared
5	Pfund	Infrared
6	Humphreys	Infrared
7+	—	Infrared / Microwave

Note: For transitions between energy levels, emission occurs when an electron moves from a higher level to a lower one ($n \to m$), while absorption occurs when it moves from a lower level to a higher one ($m \to n$).

Electrons *emit* a photon when dropping energy levels and *absorb* a photon when gaining energy levels. Since they emit and absorb light, we can relate frequency to wavelength using the speed of light:

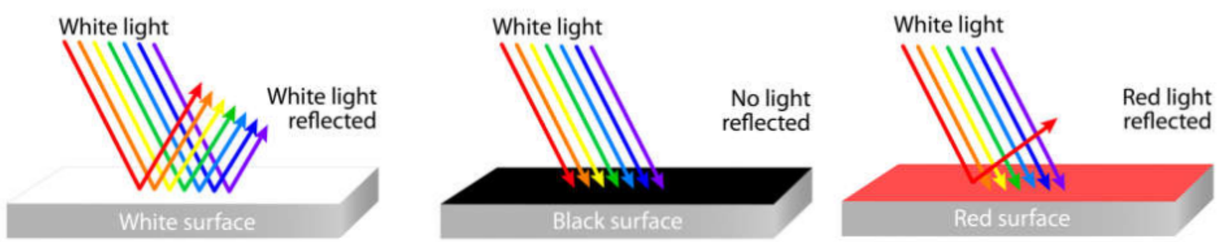
$$\boxed{c = \lambda f}$$

Blackbody Radiation

Blackbody radiation is another example of a where classical models fail to match experimental results. Notably, we will see that the classical model of blackbody radiation, the Rayleigh-Jeans law, predicts an infinite amount of energy emitted at short wavelengths (high frequencies), which is known as the *ultraviolet catastrophe*.

Thermal Radiation

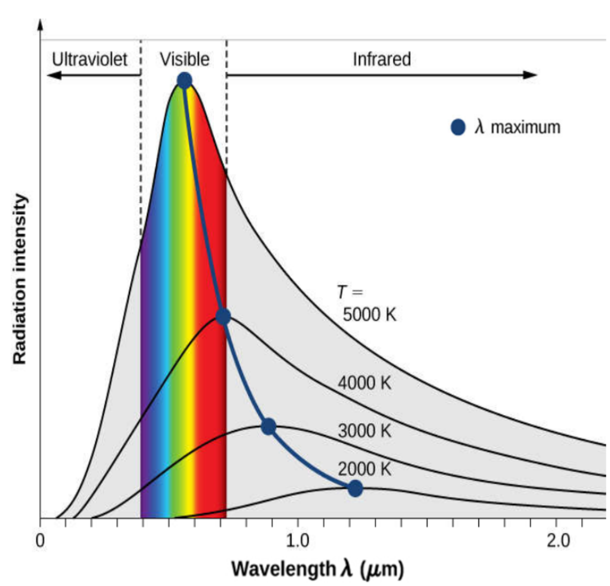
Unlike individual elements, most solids and liquids emit and absorb a continuous spectrum of radiation called **thermal radiation**, which depends on the temperature of the object. However, thermal radiation is typically not in the visible range, which is why most visible light we see from objects is reflected light. Colors result from the wavelengths of light that are reflected by an object, while the rest are absorbed.



At sufficiently high temperatures, an object begins to emit visible light as well. For example, glowing hot metal and incandescent light bulbs emit visible light due to their high temperatures.

Blackbody Radiation (Ideal Model)

To study thermal radiation without having to account for the reflected light from an object, we use the idealized model of a **blackbody**, which is an object that perfectly absorbs all incident light (zero reflection) and emits thermal radiation based on its temperature.



Because thermal radiation is dependent only on temperature, blackbodies at the *same temperature* all emit the same continuous spectrum of light, regardless of their material.

Stefan-Boltzmann finds that increasing the temperature of a blackbody increases the *intensity of all wavelengths emitted.* This is described by the **Stefan-Boltzmann Law**, which was found experimentally, that describes the intensity of radiation emitted by a blackbody at a given temperature:

$$\boxed{\frac{P}{A} = \sigma T^4, \quad \sigma = 5.67 \times 10^{-8} \, \frac{\mathrm{W}}{\mathrm{m}^2 \cdot \mathrm{K}^4}}$$

Where σ is the Stefan-Boltzmann constant, P is the power emitted by the blackbody, and A is the surface area of the blackbody.

Note: Intensity is defined as the power per unit area, which is the amount of energy emitted per second per square meter.

Additionally, the spectrum shifts towards shorter wavelengths (higher frequencies) as the temperature increases. This is described by **Wein's Law**, which was found experimentally, that describes the wavelength of peak intensity at a given temperature:

$$\boxed{\lambda_{\text{peak}} = \frac{[2.90 \times 10^6 \, \mathrm{nm} \cdot \mathrm{K}]}{T}}$$

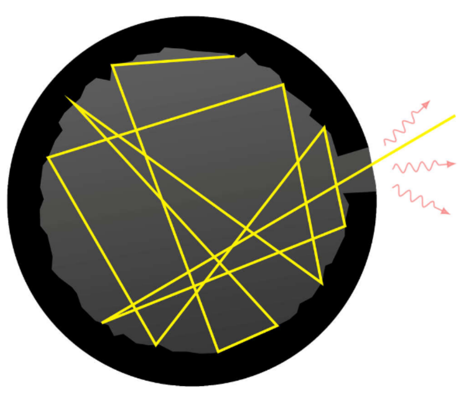
Note: Use kelvin for temperature! $(K =^\circ C + 273.15)$

Notably, the Sun is a blackbody that has the perfect temperature to emit its peak intensity in the visible range, which is why we see a continuous visible spectrum of light from the Sun.

Rayleigh-Jean Model (Classical Model)

Note (New notation): ν denotes frequency, replacing the earlier use of f.

Classically, we can model a blackbody as a cavity with perfectly reflecting walls and a small window.



Any light incident on the window it will be bounced around the reflective walls so many times that it will eventually be absorbed before it can exit the window (perfect absorption). As the walls absorb the kinetic energy of the light, it will increase in temperature.

As temperature (averaged out kinetic energy) increases to a high enough level, charges within the walls will begin to accelerate and emit light. Thus, the blackbody will emit thermal radiation.

The Rayleigh-Jean model is based on the idea that the "trapped" light inside the cavity exists as 3D standing electromagnetic waves (all trapped waves must have nodes at the walls). We can model the number of *modes per unit volume* ($N(\nu)$) of the trapped light as a function of frequency ν:

$$N(\nu) = \frac{8\pi\nu^2}{c^3}$$

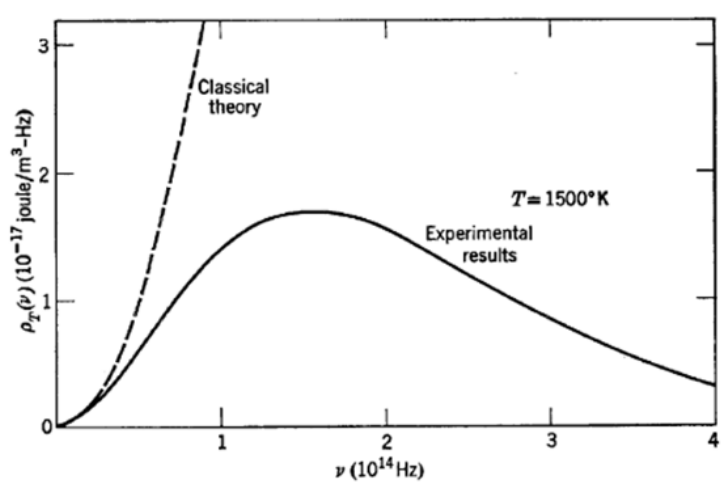
A mode refers the specific standing wave pattern that exists at a given frequency within the cavity. Because higher frequencies have shorter wavelengths, there area greater number of mode combinations (n_x, n_y, n_z) that can fit within the cavity, compared to lower frequencies.

Finally, we assign energy to each mode of the trapped light based on the equipartition theorem, giving it an average energy of $\langle E \rangle = k_B T$ per mode, where k_B is the Boltzmann constant.

Thus, the **Rayleigh-Jean model** relates the energy density ($u_T(\nu)$) contained per unit volume of the radiation within the cavity at a given frequency ν and temperature T as:

$$\boxed{u_T(\nu) = \langle E \rangle N(\nu) = \frac{8\pi\nu^2 k_B T}{c^3}, \quad k_B = 1.3806 \times 10^{-23}\, \frac{\mathrm{J}}{\mathrm{K}}}$$

Problems with the classical model:



- UV Catastrophe: The energy density goes to infinity as frequency increases, which deviates from experimental results.

- Infinite total energy: The total energy density across all frequencies should be finite but the Rayleigh-Jean model predicts an infinite total energy density emitted by the blackbody.
$$U(T) = \int_0^\infty u_T(\nu) d\nu$$

These problems are the result of our assumptions that the atoms within the wall can absorb and emit a continuous range of energies (it is actually discrete), and that every mode of oscillation is equally likely to be excited (it is actually less likely for higher frequencies).

Planck's Model (Quantized Model)

To address the inaccurate assumptions of the Rayleigh-Jean model, Max Planck proposed a new model of blackbody radiation that incorporates the idea of quantization.

He suggests that the oscillators in the walls of the cavity can only absorb and emit *discrete* packets of energy, which he called quanta. He then proposes that oscillators of frequency ν can only absorb quanta of energy *proportional to* ν:

$$E_n = nh\nu, \quad n = 1, 2, 3, \ldots$$

Where $h = 6.626 \times 10^{-34}$ J $\cdot$ s is the **Planck constant**.

This way, exciting higher frequency modes will require more energy, which is less likely to occur at a given temperature. Thus, instead of assigning an equal average energy, Planck finds:
$$\langle E \rangle = \frac{h\nu}{\exp\left(\frac{h\nu}{k_B T}\right) - 1}$$

Planck's model then states:

$$\boxed{u_T(\nu) = \frac{8\pi h}{c^3} \frac{\nu^3}{\exp\left(\frac{h\nu}{k_B T}\right) - 1}}$$

This agrees with experimental results and matches the Rayleigh-Jean model at low frequencies. Additionally, the total energy per unit volume is finite:

$$U(T) = \frac{8\pi h}{c^3} \int_0^\infty \frac{\nu^3}{e^{\frac{h\nu}{kT}} - 1} d\nu = aT^4$$

Here, we obtain a huge idea: the fact that energy is quantized. This same intuition will be applied to the atomic model, where we will see that elements emit and absorb light at discrete wavelengths because electrons can only exist at discrete energy levels.

Albert Einstein will apply this idea of quanta to light itself in the photoelectric effect, introducing the concept of photons, which are quanta of light.

Photoelectric Effect

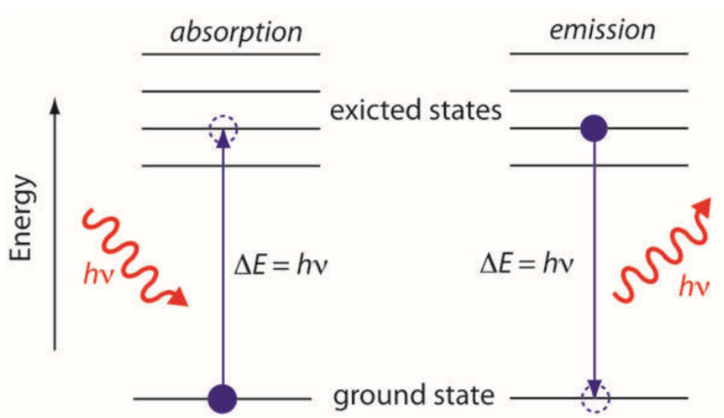
Experimentally, in 1902, Lenard established the following relationships between light incident on a metal surface and the photoelectrons emitted from that surface:

- Photoelectrons are only emitted if the frequency of the incident light is above a certain threshold frequency (ν_0) that is specific to the metal.
- Increasing the *intensity* of the incident light increases the number of photoelectrons emitted but does not affect their kinetic energy.
- Increasing the *frequency* of the incident light increases the kinetic energy (speed) of the emitted photoelectrons but does not affect the number of photoelectrons emitted.

In 1905, Einstein applies the idea of quantization to light, proposing that light is not a continuous wave but rather discrete packets of energy called **photons**. Each photon has an energy proportional to its frequency:

$$\boxed{E_{\text{photon}} = h\nu}$$

Since photons have quantized energy, they can only be absorbed by a material in integer values (it must absorb a whole photon, not a fraction of a photon). Additionally, when a substance absorbs a photon, all the photon's energy goes to *one electron*, raising its energy level.



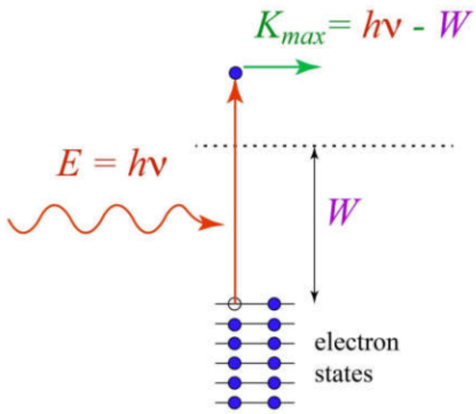
Photoelectric Effect
The photoelectric effect describes the emission (release) of electrons from a material when it absorbs photons. Thus, we describe the *minimum energy* required to emit a photoelectron as the **work function** (W) of the material.

$$\boxed{W = h\nu_0}$$

Where ν_0 is the threshold frequency of the material and is dependent on the material's properties.

If the energy of the incident photon ($E_{\text{photon}} = h\nu$) is greater than the work function, then the excess energy will be converted into kinetic energy of the emitted photoelectron:

$$\boxed{K_{\max} = h\nu - W}$$

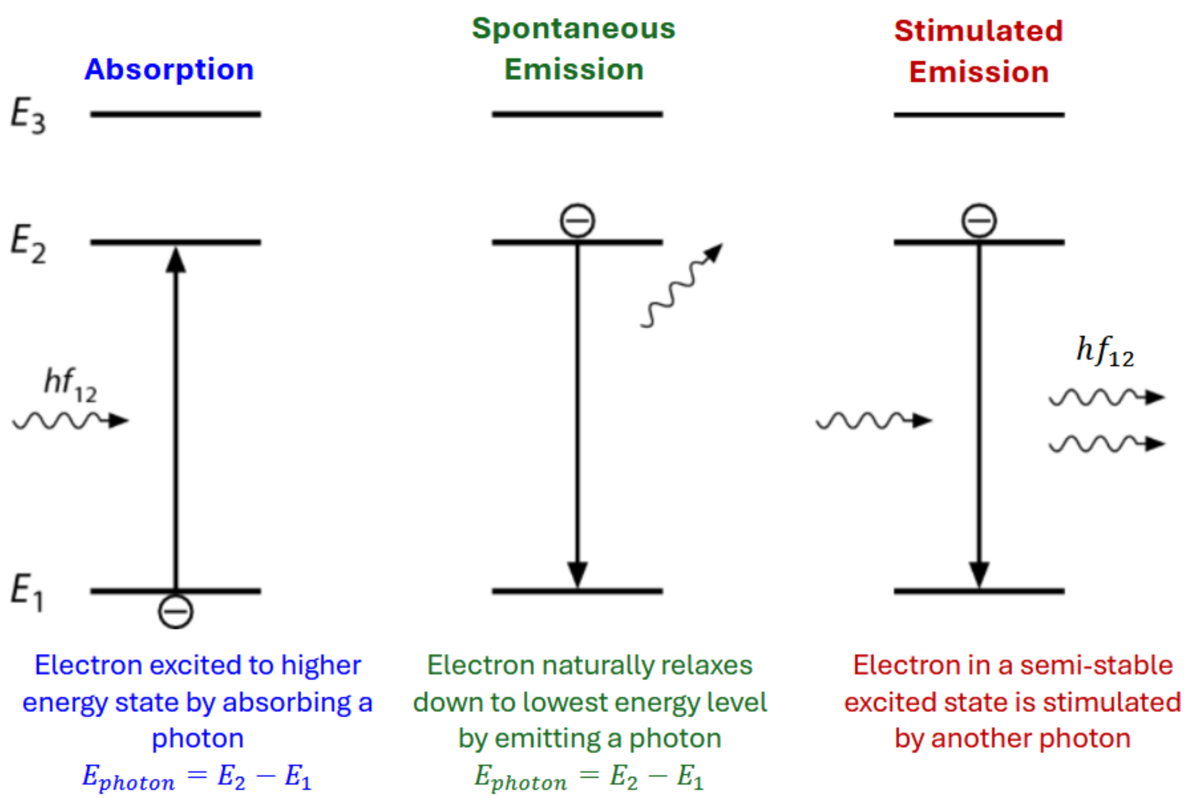


Thus, staying consistent with experimental findings:

- Incident light with higher intensity $\implies$ more incident photons and thus more photoelectrons emitted.
- Incident light with higher frequency $\implies$ higher E_{photon} and thus higher kinetic energy of emitted photoelectrons.

Electron Energy Transitions

This theory of quantized energy (photons) can also be applied to the raising and lowering of electrons between energy levels in an atom.

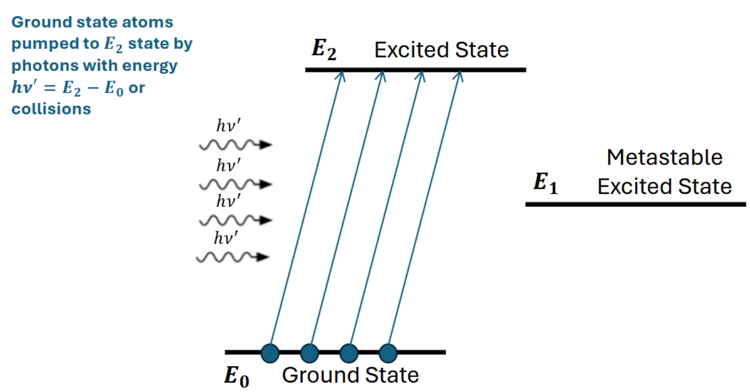


Note: Stimulated emission occurs when an incident photon of energy equal to the energy difference between two energy levels (ex: hf_{12}) causes an excited electron to drop one energy level, emitting a photon of the same energy. As a result, there will be two photons of the same energy (the original photon and the emitted photon). The original photon is not absorbed but rather just causes the electron to drop energy levels.

Lasers

A key property of stimulated emission is that the emitted photon is completely in phase with the incident photon, meaning they have the same frequency, wavelength, and direction. Thus, in a three-level laser, we can create an intense beam of coherent light by using stimulated emission.

(1)



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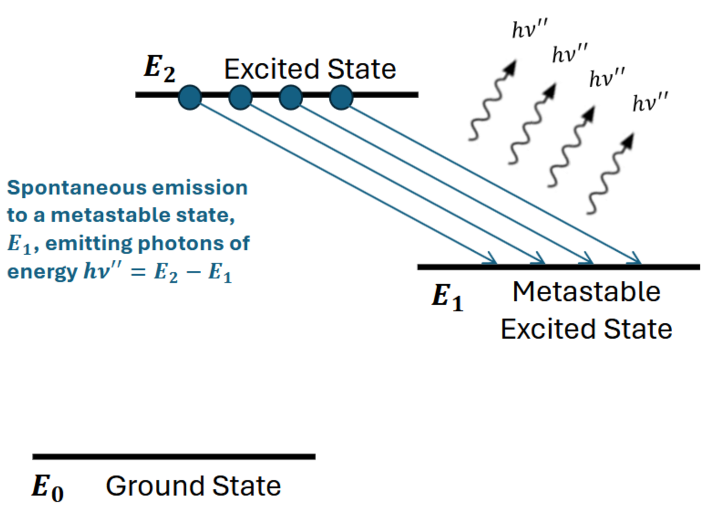
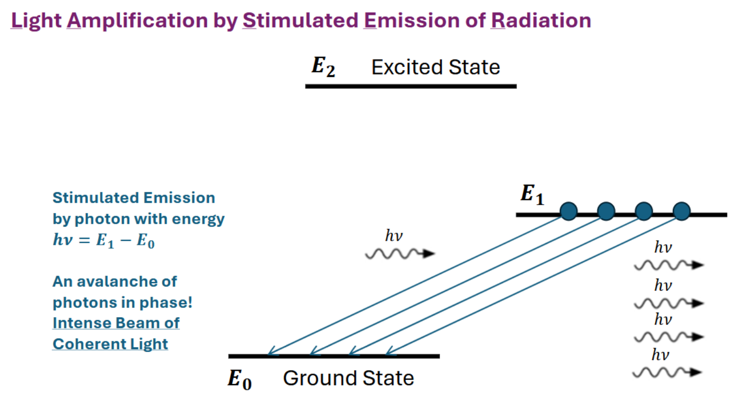


Once the electrons are at E_1, the metastable excited state, they can be held for long periods of time, waiting for the laser to be turned on.

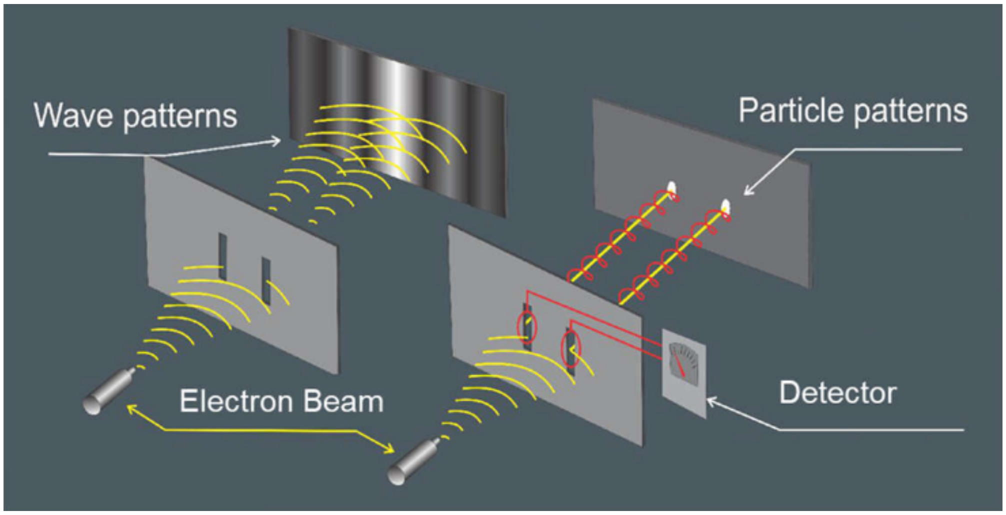
(3)



Wave-Particle Duality

One last fundamental concept of quantum mechanics is the idea of wave-particle duality, which states that all particles exhibit both wave-like and particle-like properties.

Famously, this is demonstrated by the double-slit experiment. Interestingly, photons (and electrons) exhibit both wave-like and particle-like behaviors depending on whether they are being observed. If a detector is placed at the slits to know exactly which slit the photon goes through, then the photons will behave like particles and create two bands of light on the screen. However, if there is no detector at the slits, then the photons will behave like waves and create an interference pattern on the screen.



Following the experiment with electrons, we can conclude that *all quantum particles* exhibit wave-particle duality, not just photons.

Matter Waves
De Broglie proposes that if matter particles with momentum $p = mv$ exhibit wave-like properties, then they must have a wavelength given by:

$$\boxed{\lambda = \frac{h}{p} \quad \text{(De Broglie Wavelength)}}$$

Bohr Model of the Atom

The Bohr model of the atom combines two fundamental ideas of quantum mechanics to model how electrons behave inside an atom:

- **Quantized Energy (photon):** from Planck's model of blackbody radiation and Einstein's photoelectric effect, energy transitions of electrons occur in discrete amounts:

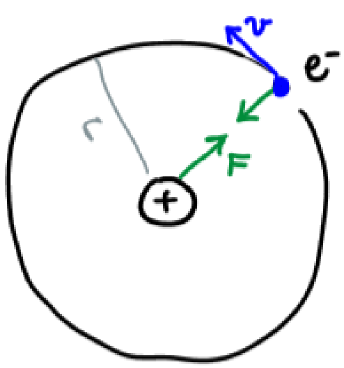
$$E_{\text{photon}} = h\nu$$

- **Wave-Particle Duality:** De Broglie's matter waves relates an electron's wavelength to its momentum

$$\lambda = \frac{h}{p}$$

Starting from the classical Rutherford model of the atom, let's first determine the energy of a bound electron in a circular orbit around the nucleus and its De Broglie wavelength.

Energy of a Bound Electron
In a hydrogen atom:



$$\sum F = m_e\vec{a}, \quad \vec{a} = \frac{v^2}{r}(-\hat{r}) \quad \text{(centripetal acceleration)}$$

$$\implies \frac{q_e q_p}{4\pi\varepsilon_0 r^2}\hat{r} = \frac{-e^2}{4\pi\varepsilon_0 r^2}\hat{r} = -m_e\frac{v^2}{r}\hat{r}$$

$$\implies v = \frac{e}{\sqrt{4\pi\varepsilon_0 m_e r}}$$

The total energy of the hydrogen atom is the sum of the kinetic energy of the electron (assumed proton at rest) and the potential energy of the electron-proton system:

$$E_H = K + U$$

$$K = \frac{1}{2}m_e v^2 = \frac{1}{2}m_e\left(\frac{e}{\sqrt{4\pi\varepsilon_0 m_e r}}\right)^2 = \frac{e^2}{8\pi\varepsilon_0 r}$$

$$U = \frac{q_1 q_2}{4\pi\varepsilon_0 r} = -\frac{e^2}{4\pi\varepsilon_0 r}$$

$$\implies \boxed{E_H = \frac{-e^2}{8\pi\varepsilon_0 r}}$$

Note that the energy of the atom depends on the radius of the electron's orbit! We will later see that this radius is quantized, which will mean energy is also quantized.

The energy is negative because the electron is bound to the nucleus, meaning it would require energy to be added to the system to free the electron from the nucleus.

De Broglie Wavelength of the Electron

Using the calculated velocity of the electron, we can find its momentum and thus its De Broglie wavelength:

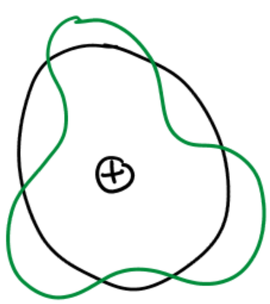
$$\lambda = \frac{h}{p} = \frac{h}{m_e v}, \quad v = \frac{e}{\sqrt{4\pi\varepsilon_0 m_e r}}$$

$$\implies \boxed{\lambda = \frac{h}{e}\sqrt{\frac{4\pi\varepsilon_0 r}{m_e}}}$$

Notice that the wavelength is also dependent on the radius of the electron's orbit.

Condition for Orbital Stability

So, what does it mean for an electron to have a wavelength? Recall that any bound wave must be a standing wave, because otherwise the wave would interfere with itself and cancel out. In the case of the bound electron, the standing wave must be bound to the *circumference* of the electron's orbit.



Thus, the *condition for orbital stability* is that the circumference of the electron's orbit must be an integer multiple of its De Broglie wavelength:

$$\boxed{2\pi r_n = n\lambda}$$

Where $n = 1, 2, 3, \ldots$ is the quantum number, and r_n is the radius of the electron's orbit containing n wavelengths.

Plugging in the expression for λ, we can find the equation for the *discrete radii* of the electron's orbit:

$$\boxed{r_n = \frac{n^2 h^2 \varepsilon_0}{\pi m_e e^2}}$$

For $n = 1$, the radius of the electron's orbit at the ground state is called the **Bohr radius**:

$$r_1 = a_B = \frac{h^2 \varepsilon_0}{\pi m_e e^2} = 5.3 \times 10^{-11}\ \text{m}$$

Using the Bohr radius, we can simplify the equation for the discrete radii of the electron's orbit:

$$\boxed{r_n = n^2 a_B}$$

Finally, using the discrete radii, we can now find the discrete energy levels of the electron in the hydrogen atom:

$$E_n = \frac{-e^2}{8\pi\varepsilon_0 r_n} = \frac{-e^2}{8\pi\varepsilon_0 n^2 a_B}$$

$$|E_1| \equiv \frac{e^2}{8\pi\varepsilon_0 a_B} = 13.6\,\text{eV} \quad \text{(ground state energy level)}$$

$$\implies \boxed{E_n = -\frac{|E_1|}{n^2}, \quad n = 1, 2, 3, \ldots}$$

At:

- $E_1 = -13.6\,\text{eV}$: ground state energy level
- $E_2, E_3, \ldots$: excited states
- $E_\infty = 0$: no longer bound to the nucleus (ionization)

Thus:

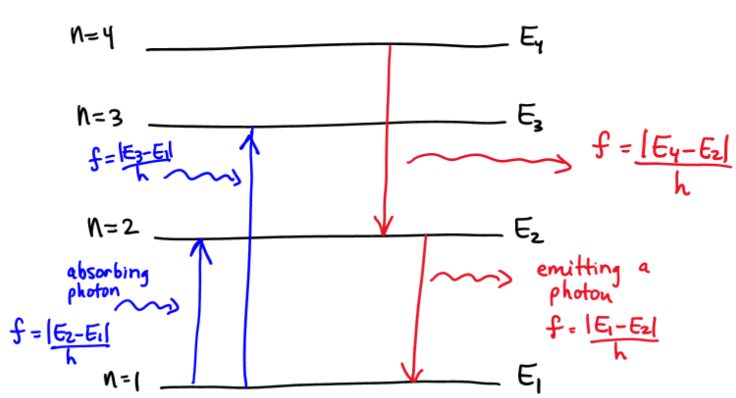
- $|E_n|$: binding energy of the electron at the nth energy level (the energy required to remove the electron from its orbit)
- E_1: ionization energy (the energy required to remove the electron from the ground state, making it an ion)

Connection to Absorption and Emission Spectra

Now that we have quantized energy levels for the electron in the hydrogen atom, we can explain the discrete wavelengths of light emitted and absorbed by hydrogen.

Since the electron absorbs a photon to move up energy levels and emits a photon to move down energy levels, the transition between energy levels must be equal to the energy of the photon:

$$E_{\text{photon}} = hf = |E_n - E_m| \implies f_{\text{photon}} = \frac{|E_n - E_m|}{h}$$



Generalizing the frequency of light absorbed or emitted for a transition between $m \to n$ energy levels:

$$\boxed{f_{m\to n} = \frac{E_n - E_m}{h} = \frac{e^2}{8\pi\varepsilon_0 h a_B}\left(\frac{1}{m^2} - \frac{1}{n^2}\right)}$$

The corresponding wavelength of light absorbed or emitted is then:

$$\lambda_{m\to n} = \frac{c}{f_{m\to n}} = \frac{8\pi\varepsilon_0 h c a_B}{e^2\left(\frac{1}{m^2} - \frac{1}{n^2}\right)}$$

$$\lambda_0 = \frac{8\pi\varepsilon_0 h c a_B}{e^2} = 91.12\,\text{nm}$$

$$\implies \boxed{\lambda_{m\to n} = \frac{\lambda_0}{(1/m^2 - 1/n^2)} \quad \text{(Balmer Formula)}}$$

Where $m = 1, 2, 3, \ldots$ and $n > m$, and λ_0 is the wavelength of transition from $m = 1$ to $n = \infty$.

Angular Momentum is also Quantized

Since the radius is quantized, the velocity is also quantized, which means the angular momentum of the electron is also quantized:

$$2\pi r_n = n\lambda = n\frac{h}{m_e v}$$

$$L = m_e v r_n = n\frac{h}{2\pi}$$

$$\implies \boxed{L = n\hbar, \quad \hbar = \frac{h}{2\pi}}$$

Note: The Bohr model works only for hydrogen and hydrogen-like ions (one electron orbiting a nucleus with atomic number Z). We account for varying nuclei with:

$$r_n = \frac{n^2 a_B}{Z}, \quad E_n = \frac{-|E_1|Z^2}{n^2}$$

Where $|E_1| = 13.6$ eV is the ground state energy level of hydrogen.

Unit 5: Quantum Mechanics

After solidifying the core concepts of quantum mechanics in unit 4, we were able to develop more accurate models of physical phenomena, such as the Bohr model of the atom. In unit 5, we will move from these conceptual foundations to introduce the formal mathematical and probabilistic framework of quantum mechanics. Shifting from deterministic classical models to probabilistic models using the wavefunction ψ.

The Wavefunction ψ

The wave-like nature of particles brings up the question of how to describe the "information" corresponding to the state of a quantum particle such as its position, momentum, energy, etc. It would turn out that because of this conflict between wave and particle properties, quantum particles *can exist in all possible states at the same time*, with a certain probability of being in each state. To better understand this, imagine a dice roll. Before the dice lands, it is in a superposition of all possible outcomes (1, 2, 3, 4, 5, 6) with some probability of landing on each outcome. It is only when the dice lands, that we *measure* the outcome, collapsing the superposition into a singular state.

The wavefunction ψ, is a mathematical *function* that contains all the probabilistic information about the state of a quantum particle. The wavefunction can be complex-valued and its magnitude squared gives the **probability density**. The probability density describes the likelihood of the particle being in a specific state. In the example below, $P(x)$ describes the probability *density* of the particle being at a specific state (position x).

Where $\psi^*(x)$ is the complex conjugate of $\psi(x)$:

$$\boxed{P(x) = |\psi(x)|^2 = \psi^*(x)\psi(x) \quad \text{(Probability Density)}}$$

Here, $P(x)$ is only the probability *density.* To find the *actual* probability of finding the particle at a specific position x within a small slice of positions δx, we need to multiply the probability density by that slice (assuming that $|\psi(x)|^2$ is approximately constant within the interval):

$$\boxed{\text{Prob(at x in } \delta x) = |\psi(x)|^2 \delta x}$$

Note that we can apply this framework to find the probability of finding the particle with a certain momentum, energy, etc. In those cases, we would use the wavefunction in momentum space or energy space, respectively. Here, $\psi(x)$ is in *position space* and therefore does not directly provide momentum or energy information.

Total Wavefunction $\mathbf{\Psi(x,t)}$

Here, I introduce the notations ψ and Ψ. The lowercase $\psi(x)$ is used to denote the *stationary* wavefunction, which is a solution to the *time-independent Schrodinger equation* and is therefore does not change with time. The uppercase $\Psi(x,t)$ is used to denote the **total wavefunction** which is a solution to the *time-dependent Schrodinger equation* and can change with time.

Intuitively, this relates to our wave-particle duality understanding. De Broglie states that any particle with momentum p has a wavelength $\lambda = h/p$. Therefore, any travelling particle must have a wavefuntion of the form of a travelling wave. Using our plane wave equation, we can write the total wavefunction of a travelling particle as:

$$\Psi(x,t) = Ae^{i(kx-\omega t)} = Ae^{ikx}e^{-i\omega t}$$

Here, we see that the total wavefunction $\Psi(x,t)$ is the product of a stationary state ($\psi(x) = A_k e^{ikx}$) and a time dependent term ($e^{-i\omega t}$).

From the De Broglie wavelength, we can relate the wavenumber k to the momentum p of the particle:

$$k = \frac{2\pi}{\lambda} = \frac{2\pi p}{h} = \frac{p}{\hbar}$$

Additionally, we can relate the angular frequency ω to the energy E_n of the particle using the quantization of energy:

$$E_n = hf = h\frac{\omega}{2\pi} = \hbar\omega$$

$$\implies \omega = \frac{E_n}{\hbar}$$

Thus, the total wavefunction of a travelling particle can be written as:

$$\boxed{\Psi(x,t) = Ae^{ipx/\hbar}e^{-iE_nt/\hbar} = \psi(x)e^{-iE_nt/\hbar}}$$

Note: This will be discussed when we introduce the time-dependent Schrodinger equation, but the total wavefunction $\Psi(x,t)$ is separable into $f(x)g(t)$ *if and only if* the potential energy of the system is independent of time.

Importantly, because the total wavefunction $\Psi(x,t)$ and the stationary wavefunction $\psi(x)$ are related by the scalar time-dependent factor $e^{-iE_nt/\hbar}$, they are *mathematically equivalent up to a complex phase shift.* This means that the amplitude of both wavefunctions are the same, and differ only by a time-varying phase angle that can be factored out or canceled during calculation.

Properties of ψ

- The Normalization Condition: The total probability of finding the particle across all space must be equal to 1 (the particle must exist somewhere):
  - Discrete case:
  $$\sum_i |\psi(x_i)|^2 \delta x = 1$$
  - Continuous case:
  $$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)\, dx = 1$$
- Finding a Specific Probability: Like discussed earlier, we must multiply the probability density ($|\psi(x_i)|^2$) by a slice of possible states (δx) to find the probability of the particle being at state x_i within that slice. Therefore, to find the probability of a particle being *within* a range of states, we just need to sum the probabilities of the particle being at each state within that range. Mathematically, this means that we sum (or integrate) the probability density over that range. Sticking with the position example:
  - Discrete case:
  $$\text{Prob(between } x_m \text{ and } x_n) = \sum_{i=m}^{n} |\psi(x_i)|^2 \delta x$$
  - Continuous case:
  $$\text{Prob(between } x_m \text{ and } x_n) = \int_{x_m}^{x_n} \psi^*(x)\psi(x)\, dx$$

  Note: This is a direct application of Prob(in δx at x).
- Finding the Expectation Value: Since we are able to find the probability of a particle being in a specific state (within a small slice of possible states), imagine the question: if we were to measure *one single particle's state many many times*, what would be the average value of those measurements? This is defined as the **expectation value** of a measurement, and we can find it by multiplying each possible state by its corresponding probability and summing (or integrating) over ALL possible states. For example, to find the expectation value of position:
  - Discrete case:
  $$\langle x \rangle = \sum_i x_i |\psi(x_i)|^2 \delta x$$
  - Continuous case*:
  $$\langle x \rangle = \int_{-\infty}^{\infty} x\psi^*(x)\psi(x)\, dx$$

  Physically, the expectation value is like the "center of mass" of the probability distribution. While the most likely state is described by the peak of the probability density, the expectation value is can tell us how *skewed* the probability distribution is.

Note: While this expectation value formulation is correct for position, for measurements such as momentum and energy, we need what we call *quantum operators* to operate on the wavefunction to find the expectation value (note that the operator and wavefunction must be in corresponding spaces).

Expectation Value of Velocity
Velocity is a derived quantity. By the Ehrenfest theorem, the expectation value of velocity can be found by

$$\langle v \rangle = \frac{d\langle x \rangle}{dt}$$

Recall that the total wavefunction $\Psi(x,t)$ has time dependence. However, the time dependent term (the phase shift) cancels out when we multiply the complex conjugates, so the following statement is equivalent, however we write it: $\Psi^*(x,t)\Psi(x,t) = \psi^*(x)\psi(x) = |\psi(x)|^2$.

Why is this important? Because of wave-particle duality, we are no longer able to predict the exact state of a particle, but rather only the probability of finding a particle in a specific state. Thus, expectation values become our best predictions! We will only know for certain the state of a particle after we measure it, which will collapse the wavefunction into a singular state.

Quantum Operators

Let's quickly finish formalizing the mathematical framework of quantum mechanics by introducing quantum operators and get to the fun part of applying this framework to solve quantum problems!!!

In the operator approach, *observable quantities* (such as position, momentum, energy, etc.) are represented by **quantum operators** that operate on the wavefunction ψ to

(1) extract possible/allowable states, and (2) the probability of being in those allowable states.

Quantum operators are defined by the following *eigenvalue problem*:

$$\boxed{\hat{A}\psi_n(x) = a_n\psi_n(x)}$$

Where:

- $\hat{A}$ is that quantum operator corresponding to the observable quantity.
- a_n is the eigenvalue that represents the allowable states of $\hat{A}$ when observed. (ex: the energy levels of an atom)
- $\psi_n(x)$ is the eigenfunction (or eigenstate) that represents the probability of being at position x and corresponds to the state a_n (eigenvalue).

Note that because the eigenfunction $\psi_n(x)$ is in position space, $|\psi_n(x)|^2$ encodes the probability density of finding the particle at position x given that the particle is in state a_n. For example, if an electron is in the second energy level E_2, then the eigenfunction $\psi_2(x)$ tells us how the electron is distributed across different positions x in space *while it stays at the second energy level.*

This is fundamentally different from the general wavefunction $\psi(x)$. While $\psi(x)$ represents a superposition of possible eigenfunctions, an eigenfunction is "pure", corresponding to a single state a_n:

$$\psi(x) = \sum_n c_n \psi_n(x)$$

Where $|c_n|^2$ represents the probability of the particle being in a particular state a_n.

Similarly, the total wavefunction is *also* a superposition of possible eigenfunctions, but with the time dependence discussed previously:

$$\Psi(x,t) = \sum_n c_n \Psi_n(x,t) = \sum_n c_n \psi_n(x) e^{-iE_n t/\hbar}$$

General Definition of Expectation Value

The expectation value of an observable quantity represented by the operator $\hat{A}$ is given by:

$$\boxed{\langle A \rangle = \int_{-\infty}^{\infty} \Psi^*(x,t) \hat{A} \Psi(x,t)\, dx}$$

Position Operator

$$\boxed{\hat{x} = x \quad \text{(Position Space)}, \quad \hat{x} = i\hbar \frac{\partial}{\partial p} \quad \text{(Momentum Space)}}$$

Momentum Operator

$$\boxed{\hat{p} = -\frac{\hbar}{i} \frac{\partial}{\partial x} \quad \text{(Position Space)}, \quad \hat{p} = p \quad \text{(Momentum Space)}}$$

This is derived directly from De Broglie's wavelength and wave-particle duality. Since a particle with momentum can be modeled as a travelling wave, we use the monochromatic plane wave (constant λ thus *constant* k) equation to derive the eigenfunction corresponding to a particle with the wavenumber k:

$$\Psi_k(x,t) = A_k e^{i(kx-\omega t)} = A_k e^{ikx} e^{-i\omega t}$$

Here, we see that the total wavefunction $\Psi_k(x,t)$ is the product of a stationary state $(\psi_k(x) = A_k e^{ikx})$ and a time dependent term $(e^{-i\omega t})$.

Looking at just the stationary state, we can find the momentum eigenfunction by applying De Broglie's wavelength to rewrite the wavenumber k in terms of momentum p:

$$\lambda = \frac{h}{p}, \quad k = \frac{2\pi}{\lambda} = \frac{2\pi p}{h} \implies p = \hbar k$$

Thus, the momentum eigenfunction is:

$$\psi_k(x) = A_k e^{ikx} \implies \psi_p(x) = A_p e^{ipx/\hbar}$$

Finally, we can use the momentum eigenfunction to derive the momentum operator by applying the eigenvalue problem:

$$\hat{p}\psi_p(x) = p\psi_p(x)$$

We are able to intuit the form of the momentum operator by looking for an operation that pulls out a constant (our eigenvalue p) while keeping the eigenfunction $\psi_p(x)$ unchanged. The only operation that does this is differentiation, which pulls out a constant $ik/\hbar$.

Note that the possible momentum values p is a *continuum* of values since the particle is a travelling wave and can have any wavelength.

Energy Operator (the Hamiltonian)

The energy operator is called the **Hamiltonian** ($\hat{H}$) and is defined as the sum of the kinetic energy operator and the potential energy operator:

$$\hat{H} = \hat{K} + \hat{U}$$

Where:

- The kinetic energy operator ($\hat{K}$) is derived from the momentum operator:

$$\hat{K} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}$$

- The potential energy operator ($\hat{U}$) is just the potential energy function $u(x)$ (typically not time dependent)

$$\hat{U} = u(x)$$

Thus, the Hamiltonian is:

$$\boxed{\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + u(x) \quad \text{(Position Space)}}$$

Remember that all of these operators can be applied to the eigenvalue problem to derive useful information about how a particle behaves. However, the eigenvalue problem for the Hamiltonian is particularly useful because of a few reasons:

- The Hamiltonian's potential energy term $u(x)$ is defined by the environment of the particle, so solving the Hamiltonian's eigenvalue problem allows us to understand how a particle behaves in a specific environment (ie. an infinite well).
- The eigenvalues of the Hamiltonian correspond to the allowable energy levels of the particle E_n. We know that these energy levels must be quantized, so solving the Hamiltonian's eigenvalue problem allows us to find the quantized energy levels of a particle in a specific environment.

Overall, this eigenvalue problem is so useful that it has its own name:

Schrodinger's Equation

Time-Independent Schrodinger's Equation (TISE)

$$\boxed{\hat{H}\psi = E\psi, \quad \hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + u(x)}$$

This is **THE most useful equation** for this unit. We will be applying this equation to four different cases (free atom, infinite well, finite well, and hydrogen atom) to find the quantized energy levels and corresponding wavefunctions of a particle in those environments.

Where:

- E is the energy eigenvalues corresponding to the quantized energy levels of the particle in that environment.
- ψ is the energy eigenfunctions where $|\psi(x)|^2$ gives the probability density of finding the particle at position x given that the particle is in energy state E.

Time-Dependent Schrodinger's Equation (TDSE)

$$\boxed{i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi}$$

This is the more general form of Schrodinger's equation. We don't use it as much to solve problems but it is important to understand that the time-independent Schrodinger's equation is just a special case of the time-dependent Schrodinger's equation where we assume that the Hamiltonian is time independent ($\hat{H}(x,t) = \hat{H}(x)$).

Additionally, if $\hat{H}(x,t) = \hat{H}(x)$, then $\Psi(x,t)$ has a unitary evolution, meaning that the time dependence of $\Psi(x,t)$ is just a complex exponential term which matches the derivation of the total wavefunction based on the plane wave equation we did earlier:

$$\Psi_n(x,t) = \psi_n(x)e^{-iE_n t/\hbar}$$

General Approach to Modeling Quantum Systems
To analyze a quantum system, we do the following:

1. Identify the energy model and define the Hamiltonian $\hat{H}$ (by knowing the potential energy function $u(x)$ of the environment).

2. Solve the time-independent Schrodinger's equation $\hat{H}\psi_n = E_n\psi_n$ to find the quantized energy levels E_n and corresponding energy eigenfunctions ψ_n.

   The TISE yields a second order differential equation. We solve this ODE by first guessing the general form of the solution based on the physical case (for example, sinusoidal solutions for the infinite square well, or exponential solutions outside a finite well with sinusoidal behavior inside), and then applying the normalization condition and the following **boundary conditions** to determine the unknown constants in the general solution:

   - ψ must be continuous
   - $\psi \to 0$ as $x \to \pm\infty$ (the particle cannot exist at infinity)
   - $\psi = 0$ where it is physically impossible for the particle to exist (ex: inside the walls of an infinite well where $u(x) = \infty$)
   - $\partial\psi/\partial x$ is continuous <u>except</u> where $u(x) = \infty$

3. Analyze the results! We can find the probability density, expectation values, etc.

Free Particle

Define the Energy Model
For a free particle, there are no external forces acting on it, so the potential energy is zero everywhere: $u(x) = 0$. Thus, it only has kinetic energy, and the Hamiltonian is:

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}$$

Set up the Time-Independent Schrodinger's Equation

$$\hat{H}\psi = E\psi \implies -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} = E\psi$$

We get the ODE:

$$\frac{\partial^2\psi}{\partial x^2} + \frac{2mE}{\hbar^2}\psi = 0$$

Let $\alpha^2 = \frac{2mE}{\hbar^2}$

$$\frac{\partial^2\psi}{\partial x^2} + \alpha^2\psi = 0$$

Solving for the Eigenvalue E
Relating α back to the energy eigenvalue E:

$$E = \frac{\hbar^2\alpha^2}{2m} \quad \text{(from rewriting } \alpha\text{)}$$

Since the total energy E of the free particle is purely kinetic, we can relate α to the wavenumber k of the particle using the De Broglie relation $p = \hbar k$:

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

$$\implies \frac{\hbar^2\alpha^2}{2m} = \frac{\hbar^2 k^2}{2m} \implies \boxed{\alpha = k}$$

Thus, the energy eigenvalues are a *continuum* of values (since k can be any value) instead of discrete values for trapped particles:

$$\boxed{E_k = \frac{\hbar^2 k^2}{2m}}$$

Physically, this makes sense since a free particle can have any momentum p and thus any energy E.

Solving for the Eigenfunction $\psi(x)$
The roots of the characteristic equation are $\pm\alpha i$, so the general solution to the ODE is:

$$\psi_k(x) = Ae^{ikx} + Be^{-ikx}$$

Note: The eigenfunctions $\psi_k(x)$ are indexed by the continuous wavenumber k, instead of discrete quantum numbers n. This is because the energy eigenvalues of a free particle are not quantized, but rather form a continuum of values. So for an eigenfunction that corresponds to a specific energy state E_k, we index it with the corresponding wavenumber k.

Applying the Normalization Condition
Consider the case where the free particle has a definite momentum, moving in the positive x direction. To determine the direction of travel, we must look at the total wavefunction $\Psi_k(x,t)$ by including the time-varying term $e^{-i\omega t}$. Recalling the monochromatic plane wave equation, the total wavefunction is expressed as:

$$\Psi_k(x,t) = Ae^{i(kx-\omega t)} + Be^{i(-kx-\omega t)}$$

Recall that the phase $(kx - \omega t)$ represents a wave moving in the positive x direction, while the phase $(-kx-\omega t)$ represents a wave moving in the negative x direction. Thus, the A term represents a wave moving in the positive x direction, while the B term represents a wave moving in the negative x direction and must be equal to zero to maintain a single definite momentum in the positive direction.

Recall: For a travelling transverse wave, if we track a single point on the wave to see its motion, the phase must be constant. Setting $kx - \omega t =$ (constant), we can see that for an increase in time t, there must be a corresponding increase in x to keep the phase constant, which means that the wave is moving in the positive x direction.

All of this to say: for a free particle with a definite momentum (single wavenumber k) in the positive x direction, the eigenfunction is:

$$\psi_k(x) = Ae^{ikx}$$

Applying the normalization condition:

$$\int_{-\infty}^{\infty} \psi_k^*(x)\psi_k(x)\,dx = \int_{-\infty}^{\infty} |A|^2\,dx = \infty \neq 1$$

Uh oh! When we attempt to normalize a free particle with a definite momentum ($\psi_k(x) = Ae^{ikx}$), we find that the total probability density is constant over all space $|\psi_k(x)|^2 = |A|^2$.

Define **Stationary State**: In quantum mechanics, a stationary state is a "pure" energy eigenfunction $\psi_k(x)$ where the resulting probability density $|\psi_k(x)|^2$ remains constant over time. Although the total eigenfunction $\Psi_k(x,t)$ for this state includes a time-dependent phase factor $e^{-iE_kt/\hbar}$, this factor cancels when calculating the probability distribution, *meaning all measurable physical observables (such as position and momentum) do not change with time.*

This normalization failure leads to two critical physical conclusions:

- Stationary States are Physically Impossible: As we know, *all stationary states* produce a constant probability density over all space (a horizontal line). Since the particle is *unbounded*, it is impossible to integrate a constant across all space and not get infinity. Therefore, the probability density ($\Psi^*\Psi$) of a free particle MUST change with time! This also makes physical sense, since we know that a free particle is moving, thus its corresponding probability density must also be moving across space.

- No Definite Momentum: Even if we included the B term in our normalization, we would still find that it equals infinity. Thus, there is no such thing as a free particle with a definite momentum ($p = \hbar k$) or energy. A physical particle *must exist as a superposition of states with different momenta*, as a single momentum state k cannot be normalized to 1.

Following these conclusions, we can conclude that the eigenfunction must then be a total eigenfunction that is a superposition of all possible energy states (wavenumbers k):

$$\Psi(x,t) = \sum_k c_k \Psi_k(x,t) = \sum_k c_k \psi_k(x) e^{-iE_k t/\hbar}$$

For a continuum of states, we replace the sum with an integral:

$$\Psi(x,t) = \int_{-\infty}^{\infty} c(k)\psi_k(x)e^{-iE_k t/\hbar} dk$$

Note: Once we accept that a free particle is a superposition of all possible momentum states, the math forces us to use the total wavefunction $\Psi(x,t)$. This is because "pure" total eigenfunctions $\Psi_k(x,t)$ with a definite momentum/energy state ALWAYS result in a stationary state (since the $e^{-iE_k t/\hbar}$ term cancels out). However, by using a superposition of total eigenfunctions $\Psi_k(x,t)$ with *different* momenta, suddenly the time dependence no longer cancels out! Basically, one conclusion always implies the other because of the math.

Finally let's solve!

We define the coefficients $c(k)$ as:

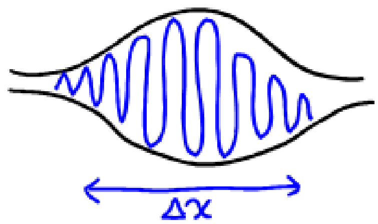
$$c(k) = \frac{\phi(k)}{\sqrt{2\pi}}$$

Where $\phi(k)$ is the *wavefunction in momentum space*, mathematically, this relates the total wavefunction in position space to the wavefunction in momentum space through a Fourier transform:

$$\boxed{\Psi(x,t) = \int_{-\infty}^{\infty} \frac{\phi(k)}{\sqrt{2\pi}}\psi_k(x)e^{-iE_k t/\hbar} dk = \int_{-\infty}^{\infty} \frac{\phi(k)}{\sqrt{2\pi}} e^{i(kx - E_k t/\hbar)} dk}$$

The Result: A "Wavepacket"
The physical result of all this math can be understood through the Fourier transform. A Fourier transform decomposes a function into its constituent frequencies, each with a different

weight (c_k). In this case, the total wavefunction $\Psi(x,t)$ is a superposition of plane waves with different wavenumbers k, each with its own weight c_k. These individual plane waves span all space, so there exists *destructive interference* as the wavenumbers vary continuously. However, there will also be a small localized region of constructive interference where the plane waves are approximately in phase, creating a "wavepacket", which corresponds to the particle!



Mathematically, the *more* individual waves (with differing k and thus differing wavelengths) we add together, the more interference we get, which leads to a more localized wavepacket (and more cancellation outside the wavepacket). Therefore, a very localized position of the particle (small Δx) means a very uncertain k and p!!! (and vice versa)

Heisenberg Uncertainty Principle
Thus, Heisenberg states that it is IMPOSSIBLE to know a particle's exact position and momentum at the same time. Importantly, this applies to not just the case of the free particle, but to ALL quantum particles!

$$\boxed{\Delta x \Delta p \geq \frac{\hbar}{2}}$$

Where:

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} \quad \text{(uncertainty of position)}$$

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} \quad \text{(uncertainty of momentum)}$$

Infinite Potential Well (Particle in a Box)

Finite Potential Well

Hydrogen Atom